

Upcoming Due Dates

- HW8 Due Tuesday, March 24
- Module 4 Worksheet – Due Tuesday, March 24
- Proficiency attempt #3 Tuesday, March 24
- HW9 – Mod 5 Prelab Due Thursday, March 26
– Rolling must be completed by Wednesday Night
- Module 4 video – Due Sunday Mar. 29

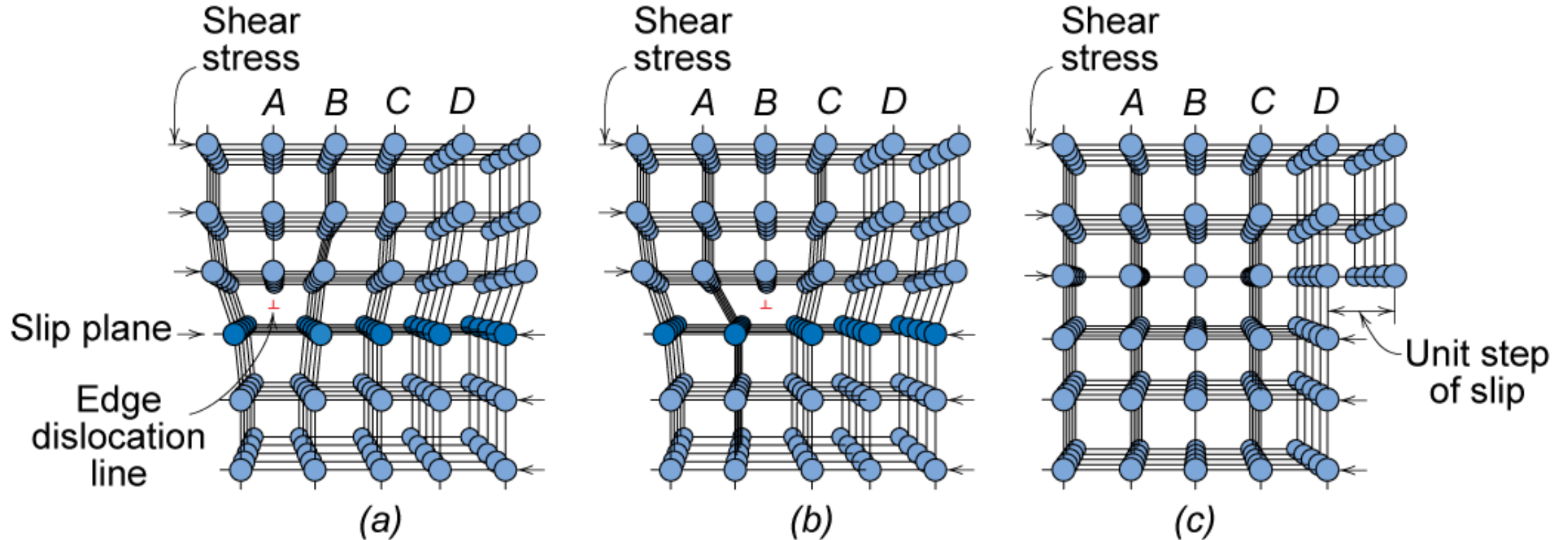
Proficiency Testing Study Session

- **JLK 323**
- **Monday, 3-23**
- **8 – 9 PM**

Chapter 8:

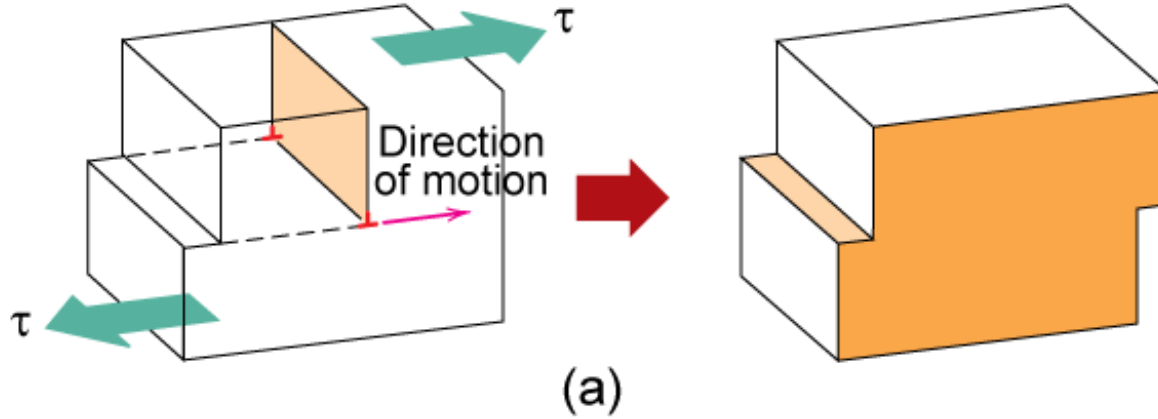
Deformation vs. Fracture

Dislocation motion & plastic deformation

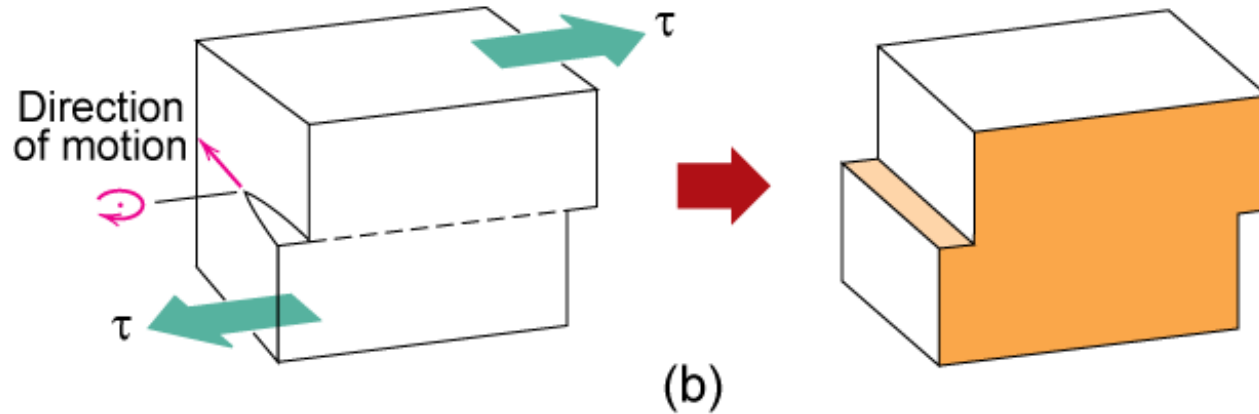


Instead of all bonds on a plane, only the bonds along dislocation line are broken

A dislocation moves along a slip plane in a slip direction perpendicular to the dislocation line.



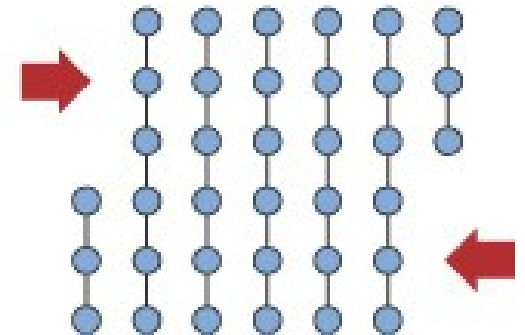
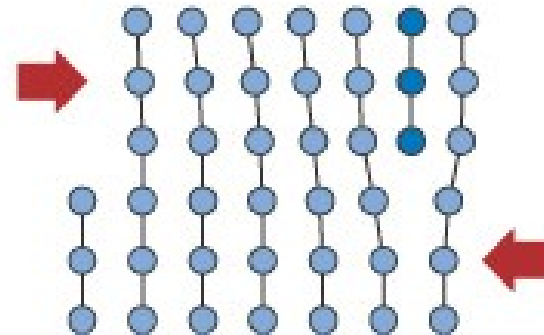
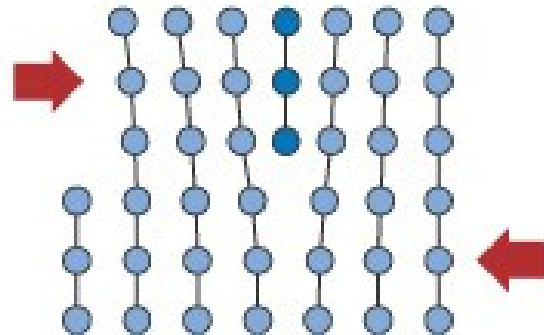
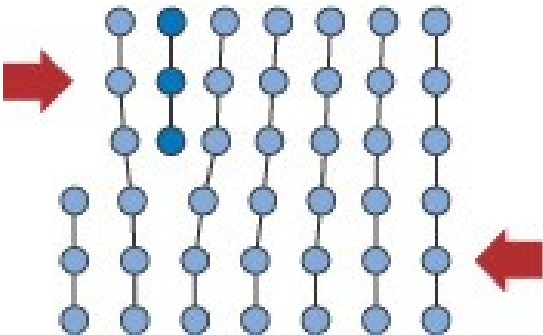
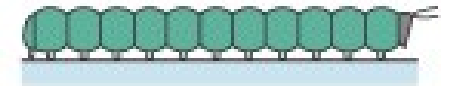
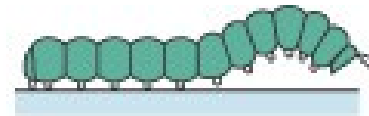
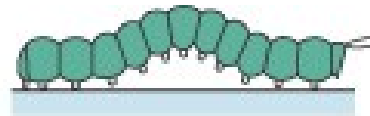
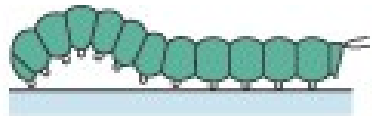
**Edge
dislocation**



**Screw
dislocation**

Representation of the analogy between caterpillar and dislocation motion

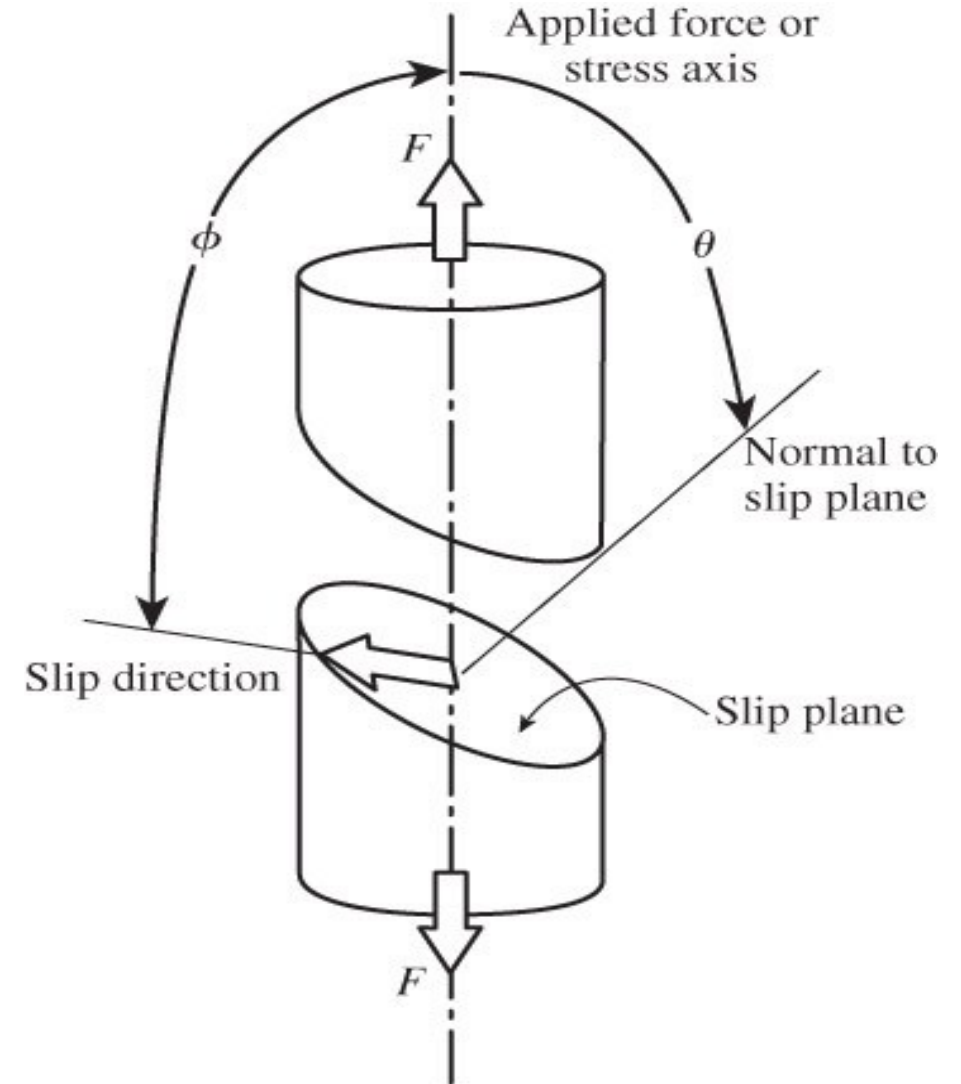
c08f03



Plastic Deformation in Single-crystal Metals: The Process of Slip

- When a force acts perpendicular (or "normal") to the surface of an object, it exerts a normal stress: σ
- When a force acts parallel to the surface of an object, it exerts a shear stress: τ

$$\tau = \sigma \cos \theta \cos \phi$$



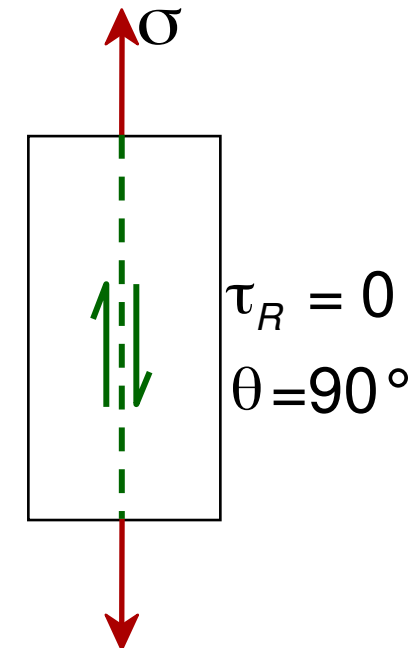
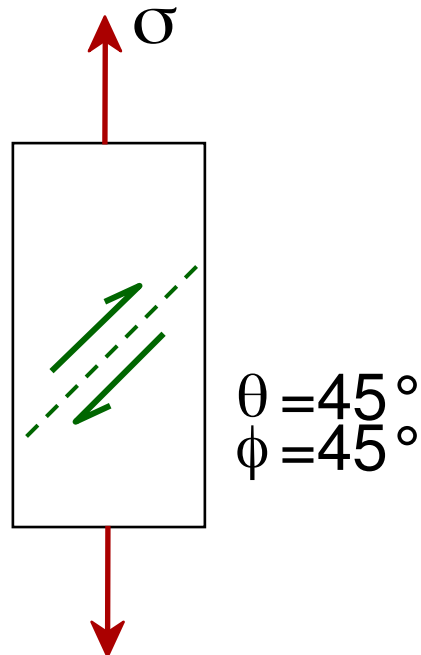
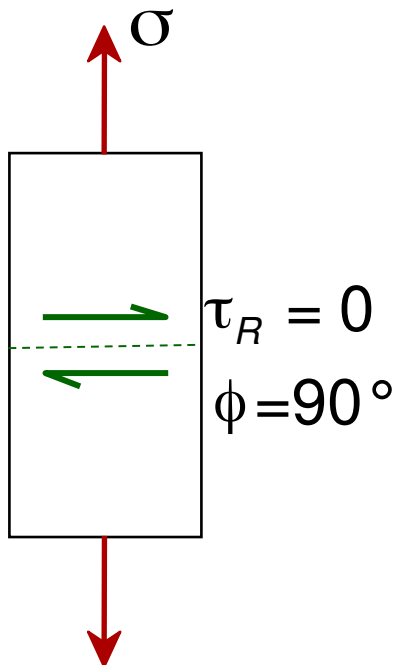
Critical Resolved Shear Stress, τ_{CRSS}

- Condition for dislocation motion:
- Ease of dislocation motion depends on crystallographic orientation

$$\tau > \tau_{CRSS}$$

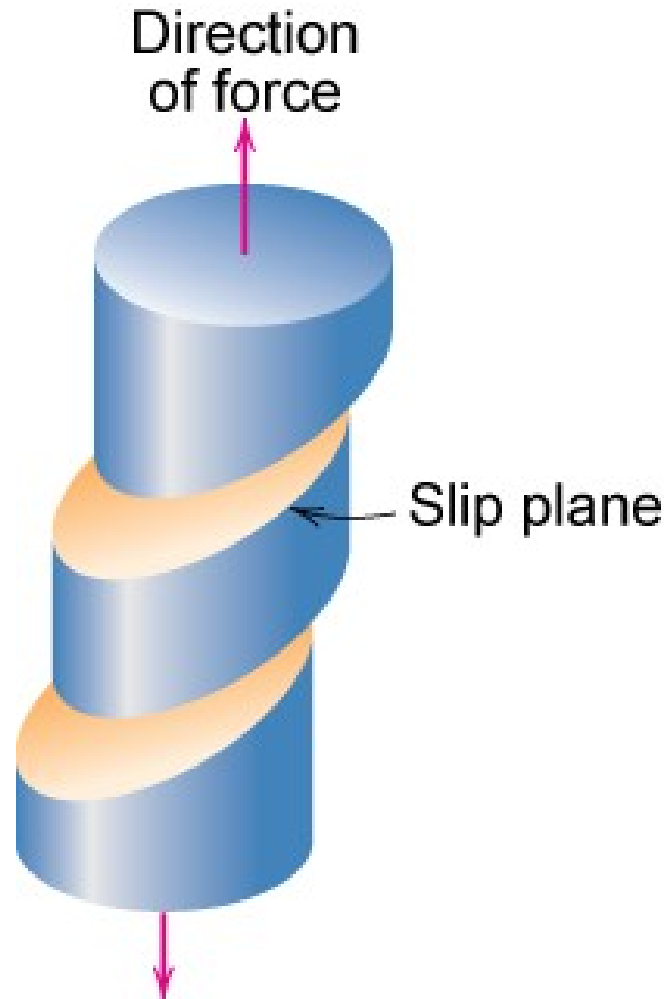
typically

10^{-4} GPa to 10^{-2} GPa



τ maximum at $\theta = \phi = 45^\circ$

Single Crystal Slip



Adapted from Fig. 8.9, *Callister & Rethwisch 3e.*

Adapted from Fig. 8.8, *Callister & Rethwisch 3e.*



Plastic Deformation in Single-crystal Metals: The Process of Slip

- Resolved shear stress

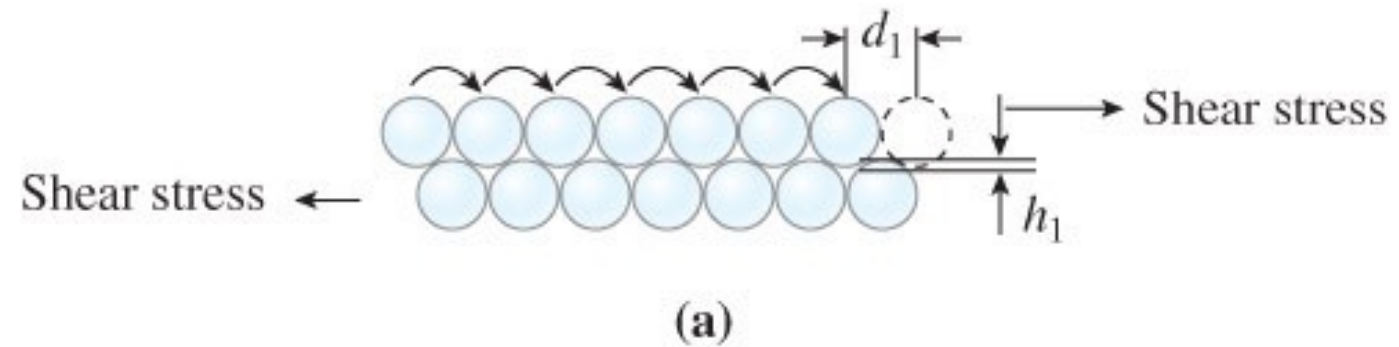
$$\tau = \sigma \cos \theta \cos \phi$$

- Critical resolved shear stress

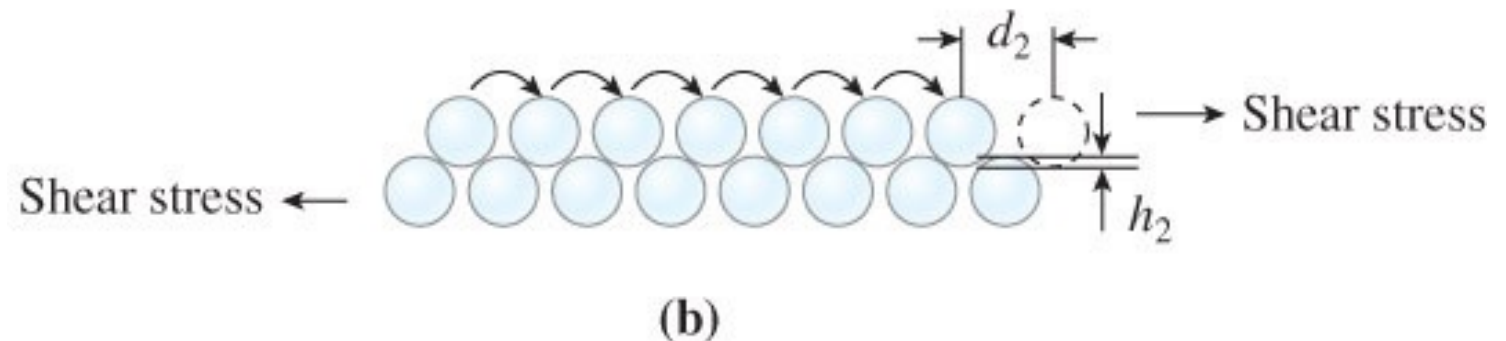
$$\tau_{\text{CRSS}} = \sigma_y \cos \theta \cos \phi$$

If the resolved shear stress is greater than critical resolved shear stress, plastic deformation occurs in metals

In real crystals, slip occurs on close-packed planes and in close-packed directions in what are known as "slip systems".



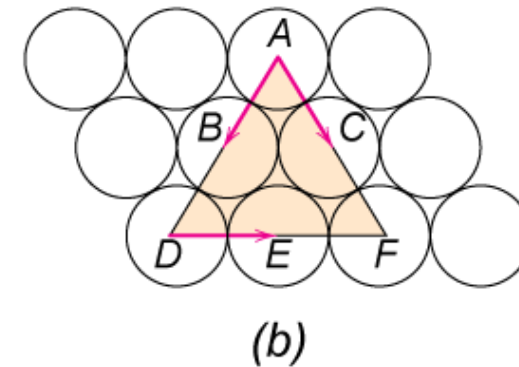
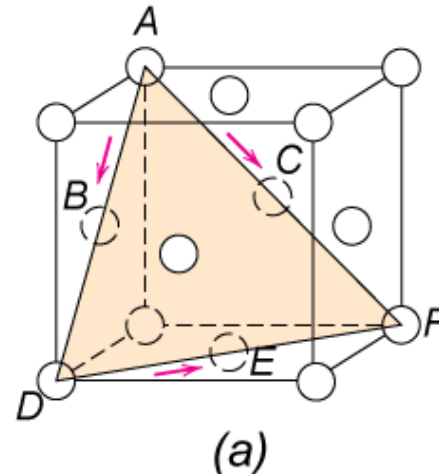
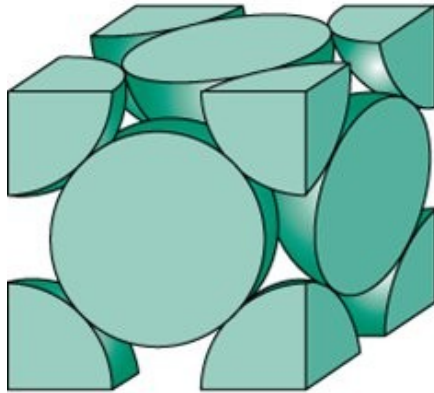
Easier to move atoms that are closer together!



Deformation Mechanisms

Slip System

- **Slip plane** - plane on which easiest slippage occurs; planes with highest planar densities (close-packed)
- **Slip direction** - direction of dislocation movement; directions with the highest linear densities (close-packed)



Adapted from Fig. 8.6, Callister & Rethwisch 3e.

- FCC Slip occurs on **{111} planes** (close-packed) in **$\langle 110 \rangle$ directions** (close-packed)
=> total of 12 slip systems in FCC
- For BCC & HCP there are other slip systems.

Slip System

- **FCC (Cu, Al, Au): 12**
- **BCC (α –iron, W): 12 or 24**
- **HCP (Ti, Zn): 3 or 6**

- **More slip systems: More ductile**

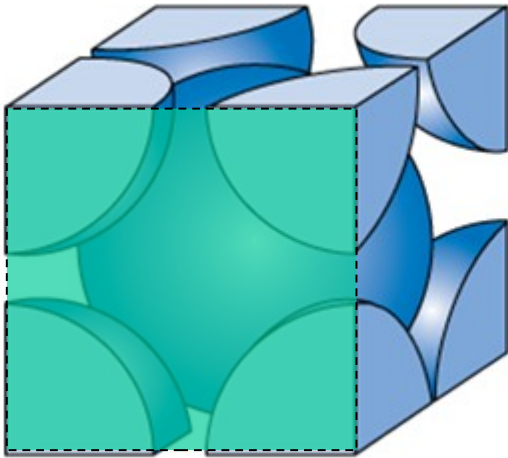
Planar Density and Linear Density

PD = (number of atoms on a plane)/(area of the plane)

LD = (number of atoms on a direction)/(length of the direction vector)

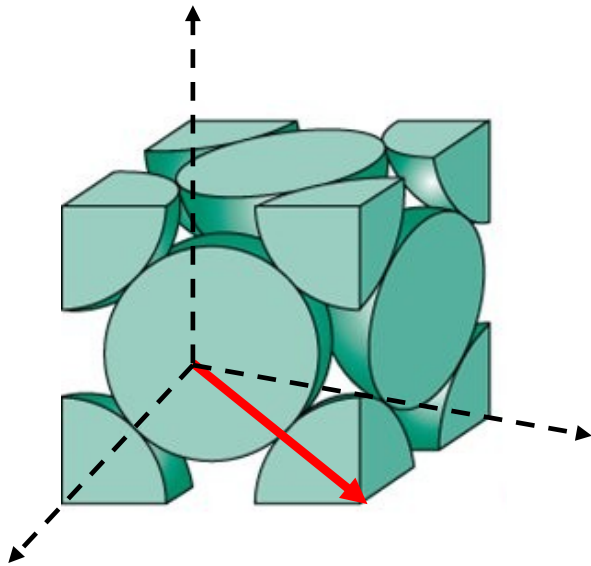
Example Calculations

1. Calculate the planar density for (100) plane for a BCC structure. Express the result in terms of edge length of the unit cell, a .



Example Calculations

- Calculate the linear density for $[110]$ direction for the FCC structure. Express the result in terms of edge length of the unit cell, a .



Slip Motion in Polycrystals

- Slip planes & directions change from one grain to another.
- τ_R will vary from one grain to another.
- The grain with the largest τ yields first.
- Other (less favorably oriented) crystals yield later.
- Stronger - grain boundaries pin deformations





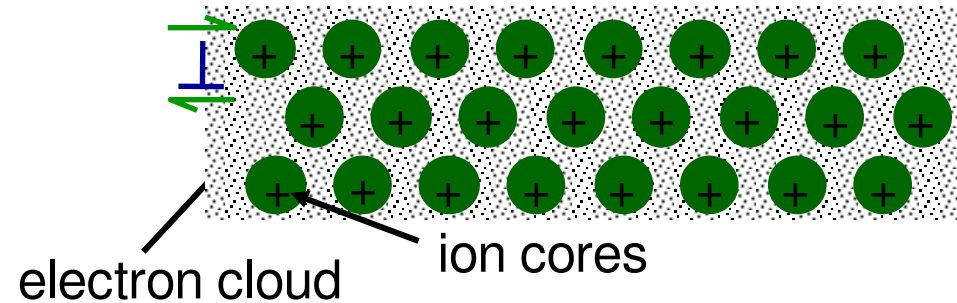
Brittle Fracture



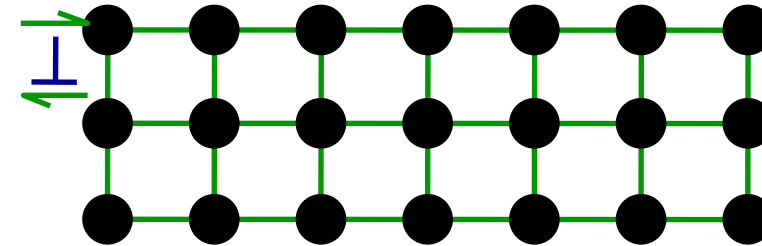
Ductile fracture
Why necking?

Here is why!

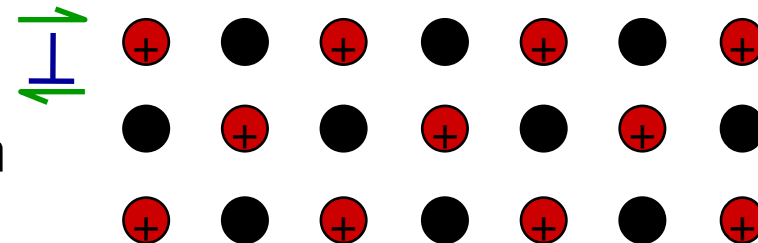
- Metals (Cu, Al):
Dislocation motion is easiest
 - non-directional bonding
 - close-packed directions for slip



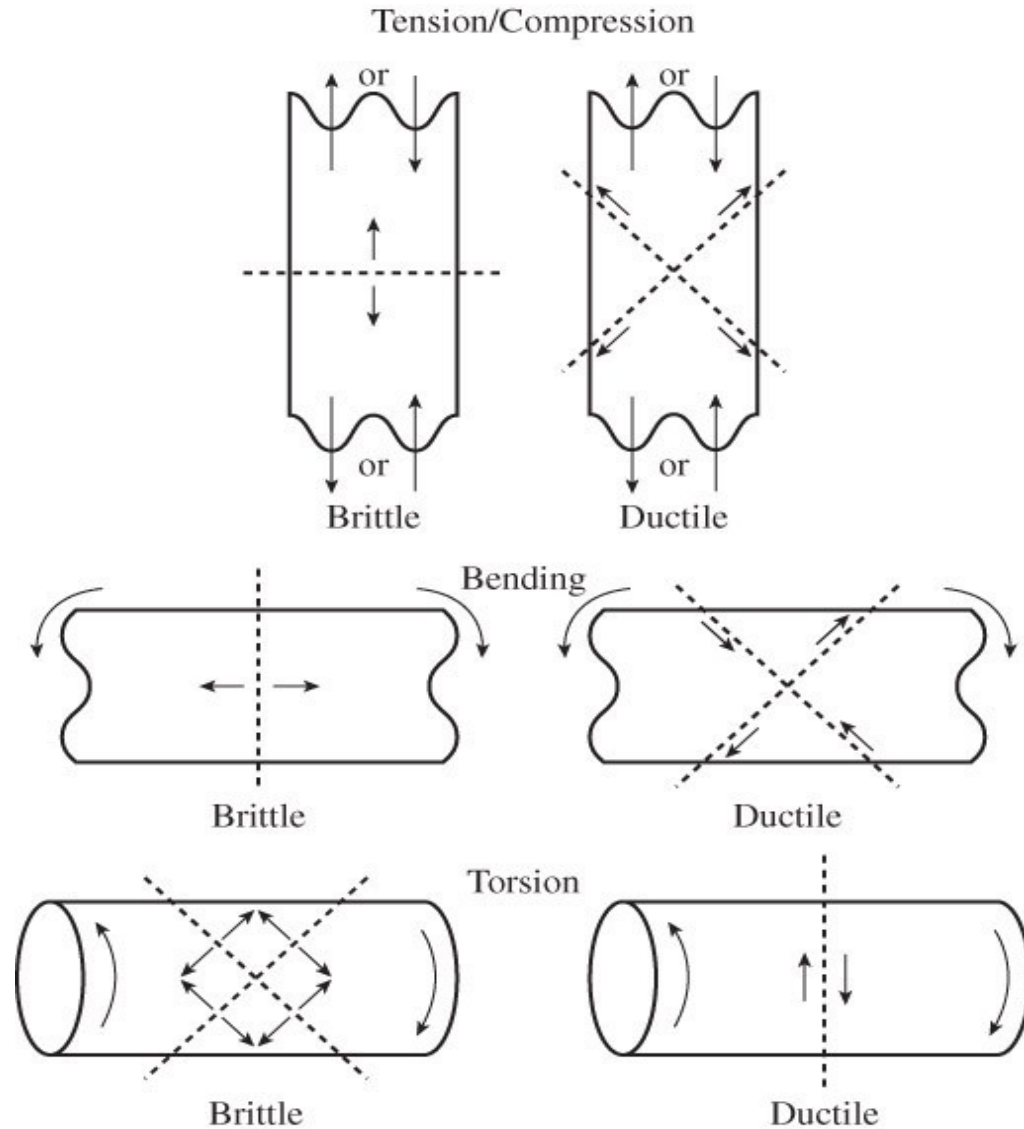
- Covalent Ceramics
(Si, diamond): Motion is difficult
 - directional (angular) bonding



- Ionic Ceramics (NaCl):
Motion is difficult
 - need to avoid nearest neighbors of like sign (- and +)

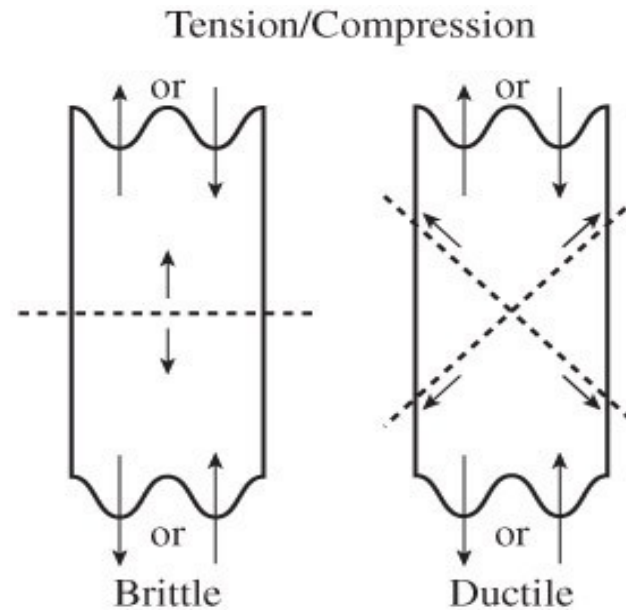
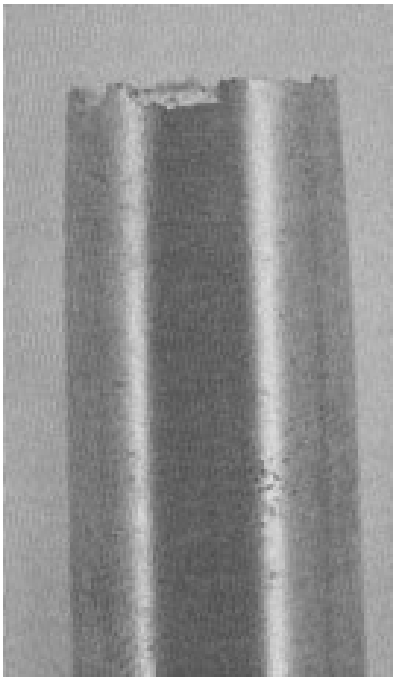
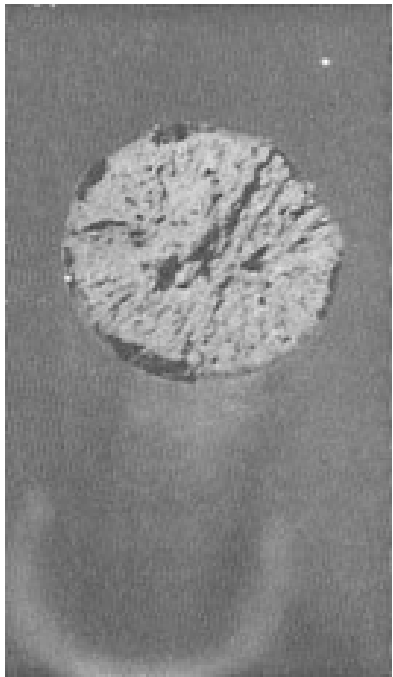


Stress on different planes under various loading condition

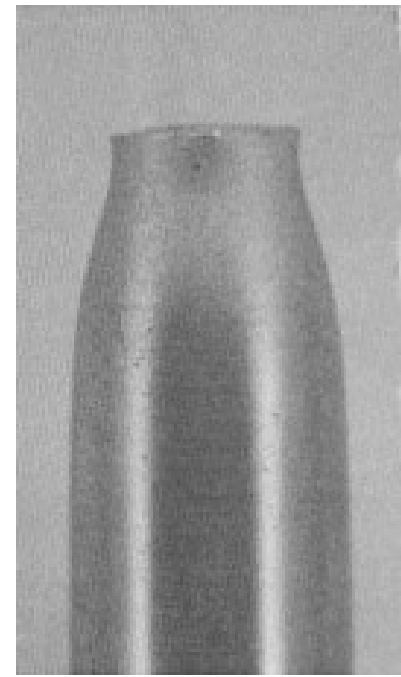
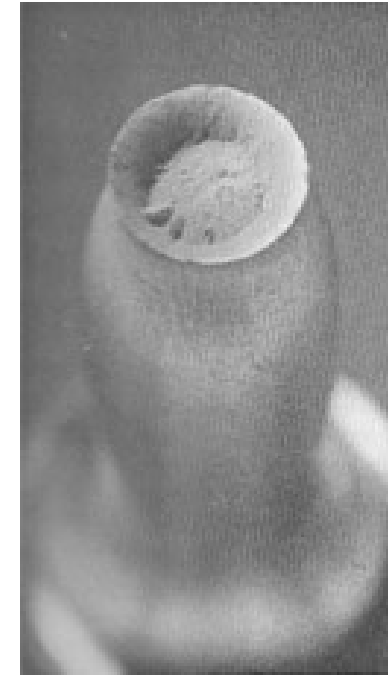


Brittle vs. Ductile Fracture: Tensile Loading

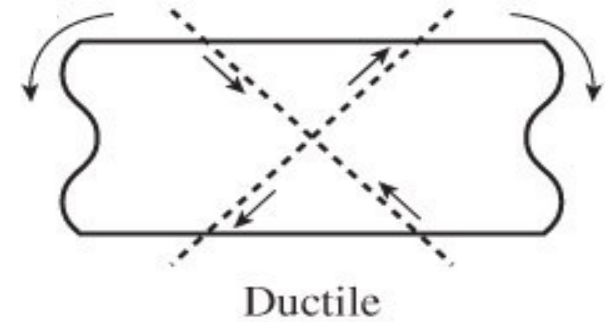
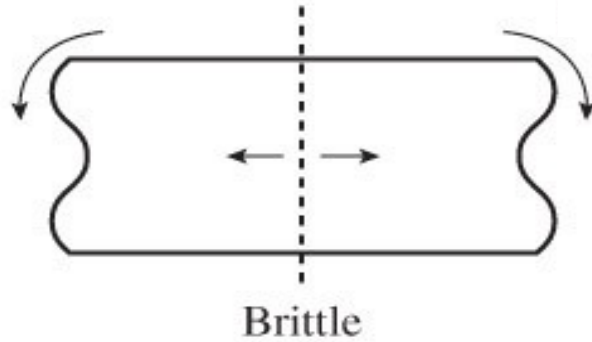
Brittle Fracture:
Fails by tension



Ductile fracture:
Fails by shear



Brittle vs. Ductile Fracture: 3-Point Bending

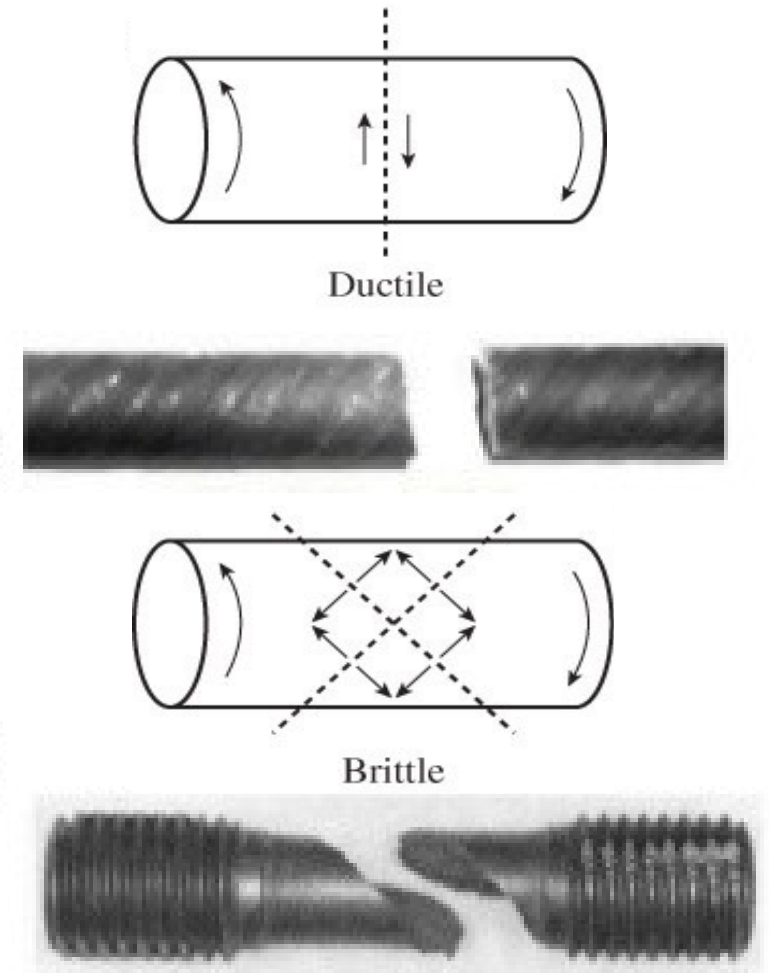


<https://www.youtube.com/watch?v=HYn816DvqSQ>

<https://www.komaspec.com/about-us/blog/guide-to-sheet-metal-bending/>

Brittle vs. Ductile Fracture: Torsional Loading

- **Ductile Materials:**
 - Generally fail in shear
 - Ductile specimen breaks along the plan of maximum shear
 - i.e. a plan perpendicular to the shaft axis
- **Brittle Materials:**
 - Weaker in tension
 - Brittle specimen breaks along planes perpendicular to the direction in which tension is maximum.
 - i.e. along surfaces at 45° to the shaft axis.



Fracture mechanisms

- Ductile fracture
 - Accompanied by significant plastic deformation
- Brittle fracture
 - Little or no plastic deformation
 - Catastrophic

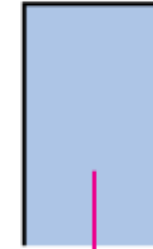
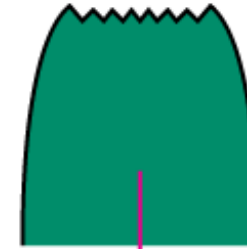
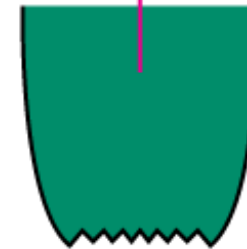
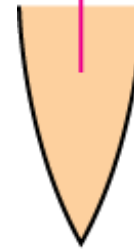
Ductile vs Brittle Failure

Fracture
behavior:

Very
Ductile

Moderately
Ductile

Brittle



Adapted from Fig. 9.1,
Callister & Rethwisch 3e.

$\%AR$ or $\%EL$

Large

Moderate

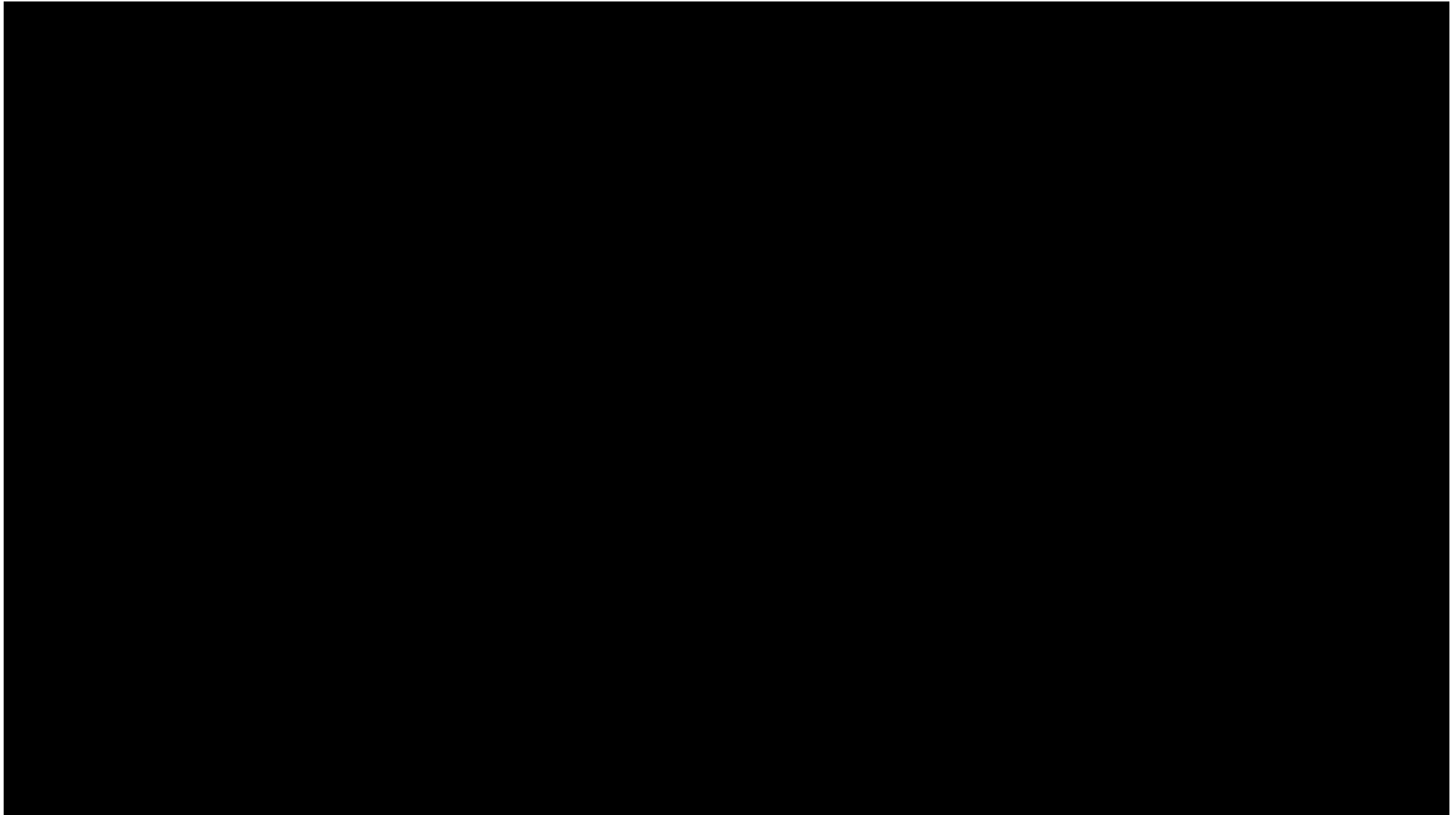
Small

- Ductile fracture is usually more desirable than brittle fracture!

Ductile:
Warning before
fracture

Brittle:
No
warning

Gold Atoms

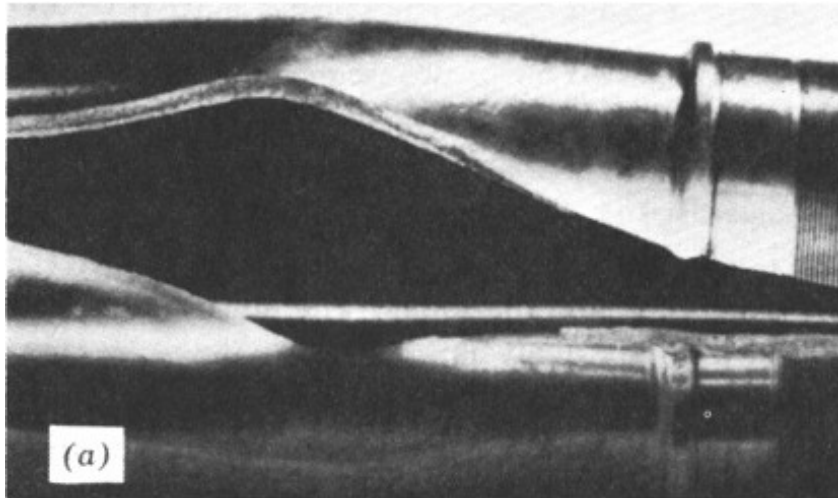


<https://www.youtube.com/watch?v=pGWSX6pStd0>

Example: Pipe Failures

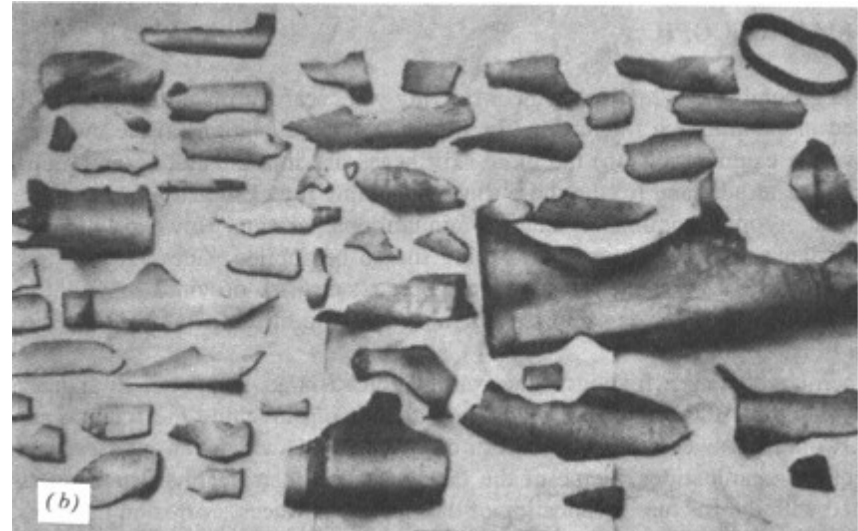
- **Ductile failure:**

- one piece
- large deformation



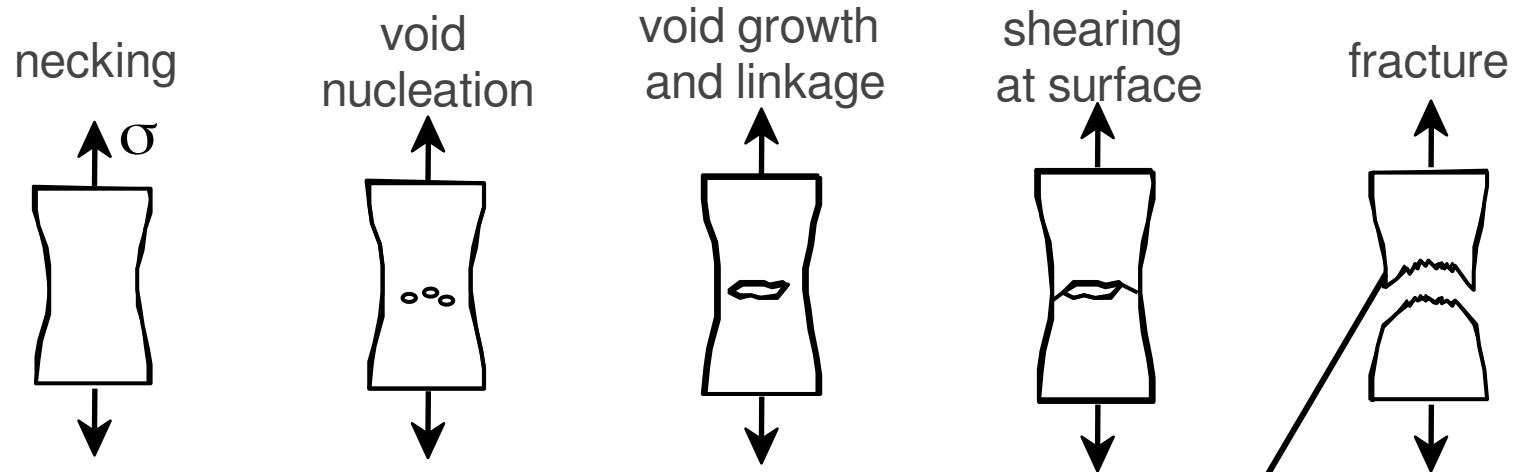
- **Brittle failure:**

- many pieces
- small deformations



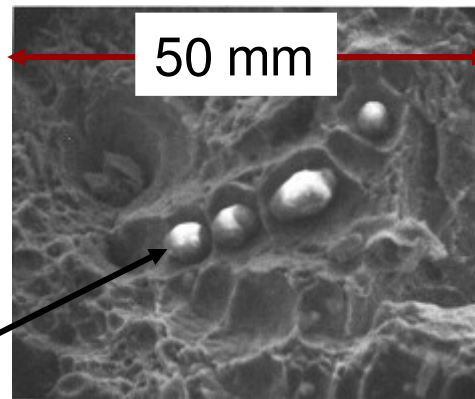
Moderately Ductile Failure

Evolution to failure:

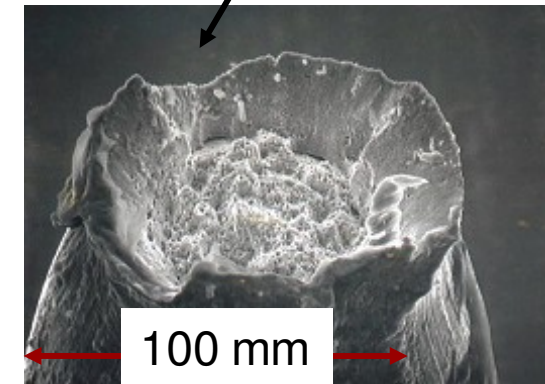


Resulting fracture surfaces
(steel)

particles
serve as void
nucleation
sites.



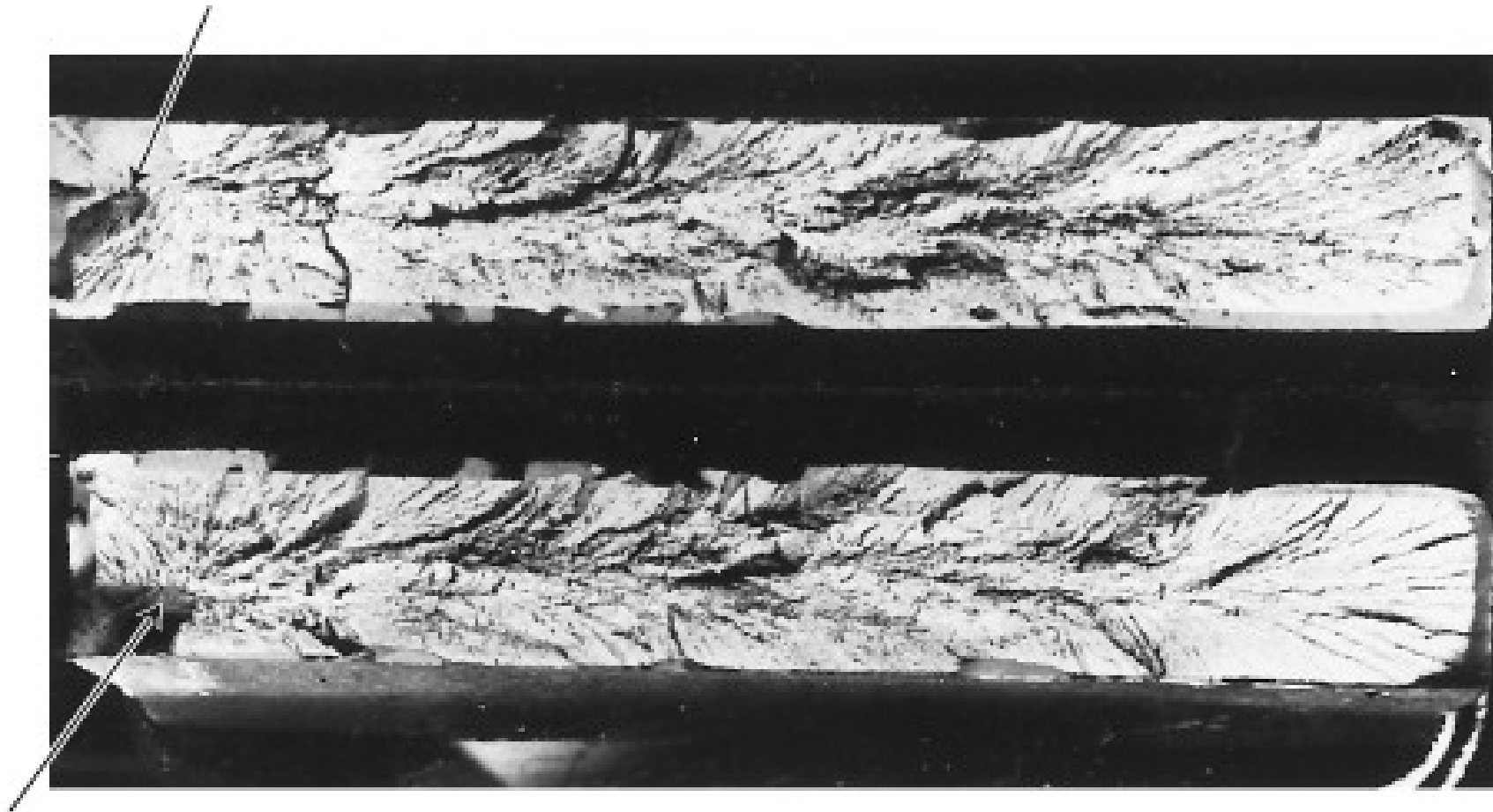
From V.J. Colangelo and F.A. Heiser, *Analysis of Metallurgical Failures* (2nd ed.), Fig. 11.28, p. 294, John Wiley and Sons, Inc., 1987. (Orig. source: P. Thornton, *J. Mater. Sci.*, Vol. 6, 1971, pp. 347-56.)



Fracture surface of tire cord wire loaded in tension. Courtesy of F. Roehrig, CC Technologies, Dublin, OH. Used with permission.

Brittle Failure

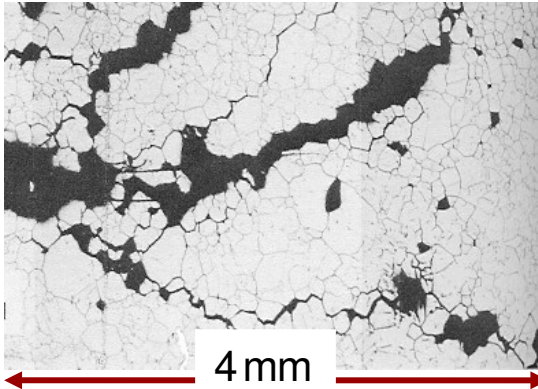
Arrows indicate point at which failure originated



Adapted from Fig. 9.5(a), *Callister & Rethwisch 3e*.

Brittle Fracture Surfaces

- Intergranular
(between grains)



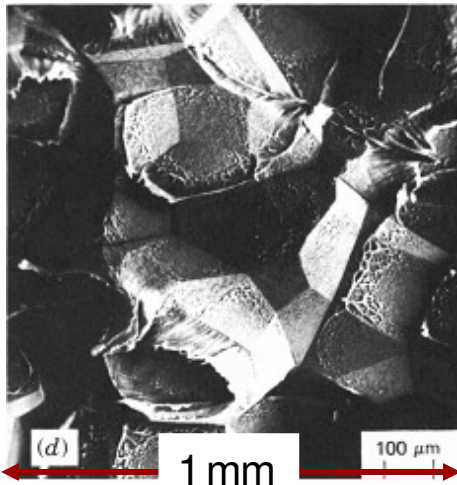
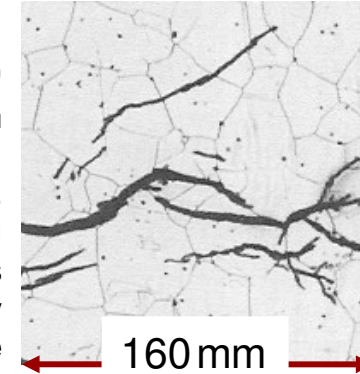
304 S. Steel (metal)

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9th ed, Fig. 633, p. 650.
Copyright 1985, ASM
International, Materials
Park, OH. (Micrograph by
J.R. Keiser and A.R.
Olsen, Oak Ridge National
Lab.)

- Intragranular
(within grains)

316 S. Steel (metal)

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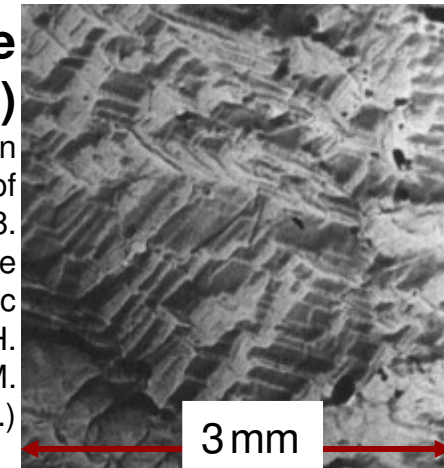


Polypropylene (polymer)

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from R.W. Hertzberg,
"Deformation and
Fracture Mechanics of
Engineering Materials",
(4th ed.) Fig. 7.35(d), p.
303, John Wiley and
Sons, Inc., 1996.

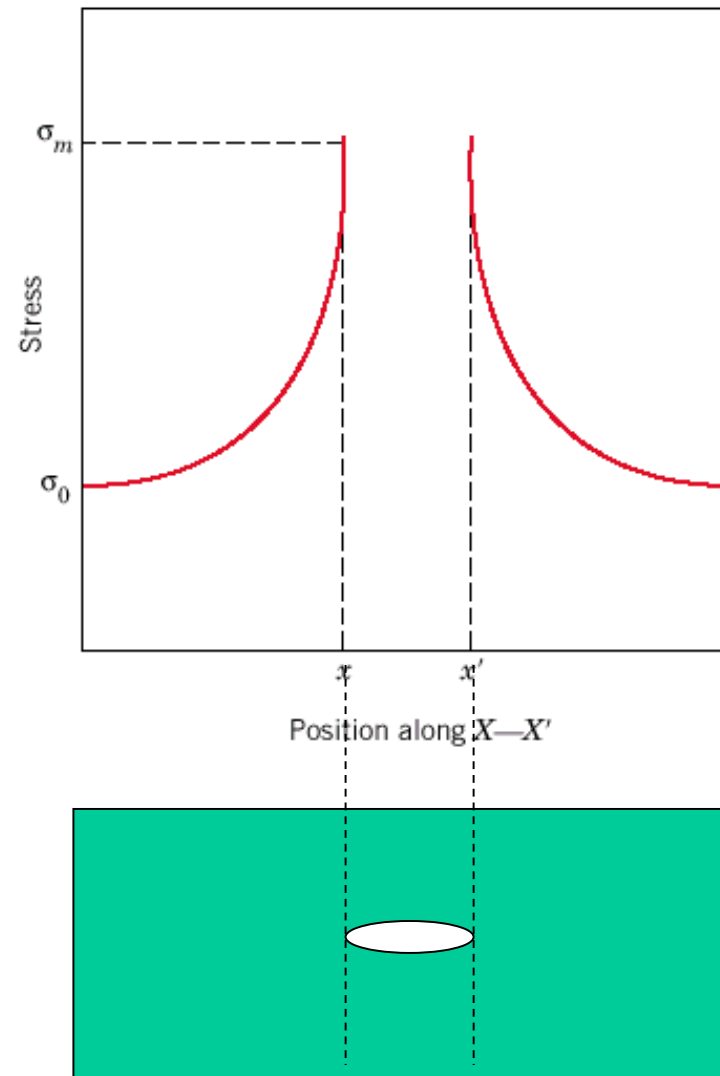
Al Oxide (ceramic)

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Brittle Materials", p. 78.
Copyright 1990, The
American Ceramic
Society, Westerville, OH.
(Micrograph by R.M.
Gruver and H. Kirchner.)



(Orig. source: K. Friedrich, *Fracture* 1977, Vol. 3,
ICF4, Waterloo, CA, 1977, p. 1119.)

Concentration of Stress at Crack Tip

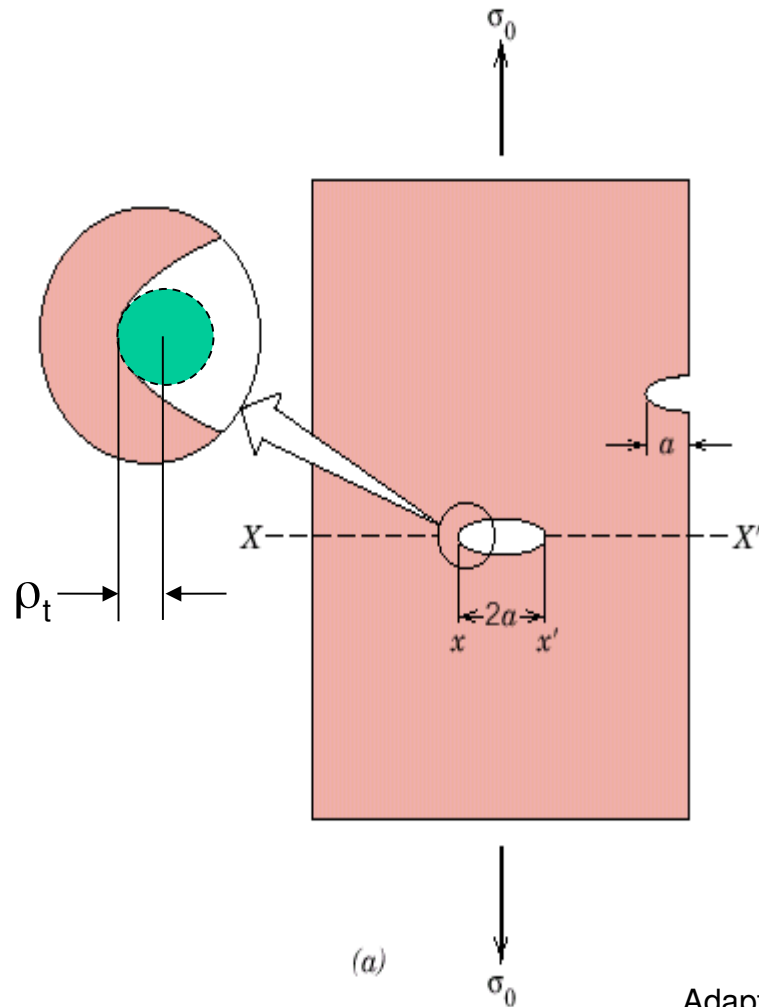


Adapted from Fig. 9.8(b),
Callister & Rethwisch 3e.

Flaws are Stress Concentrators!

Results from crack propagation

- Griffith Crack



$$\sigma_{\max} = 2\sigma_o \left(\frac{a}{\rho_t} \right)^{1/2} = K_t \sigma_o$$

where

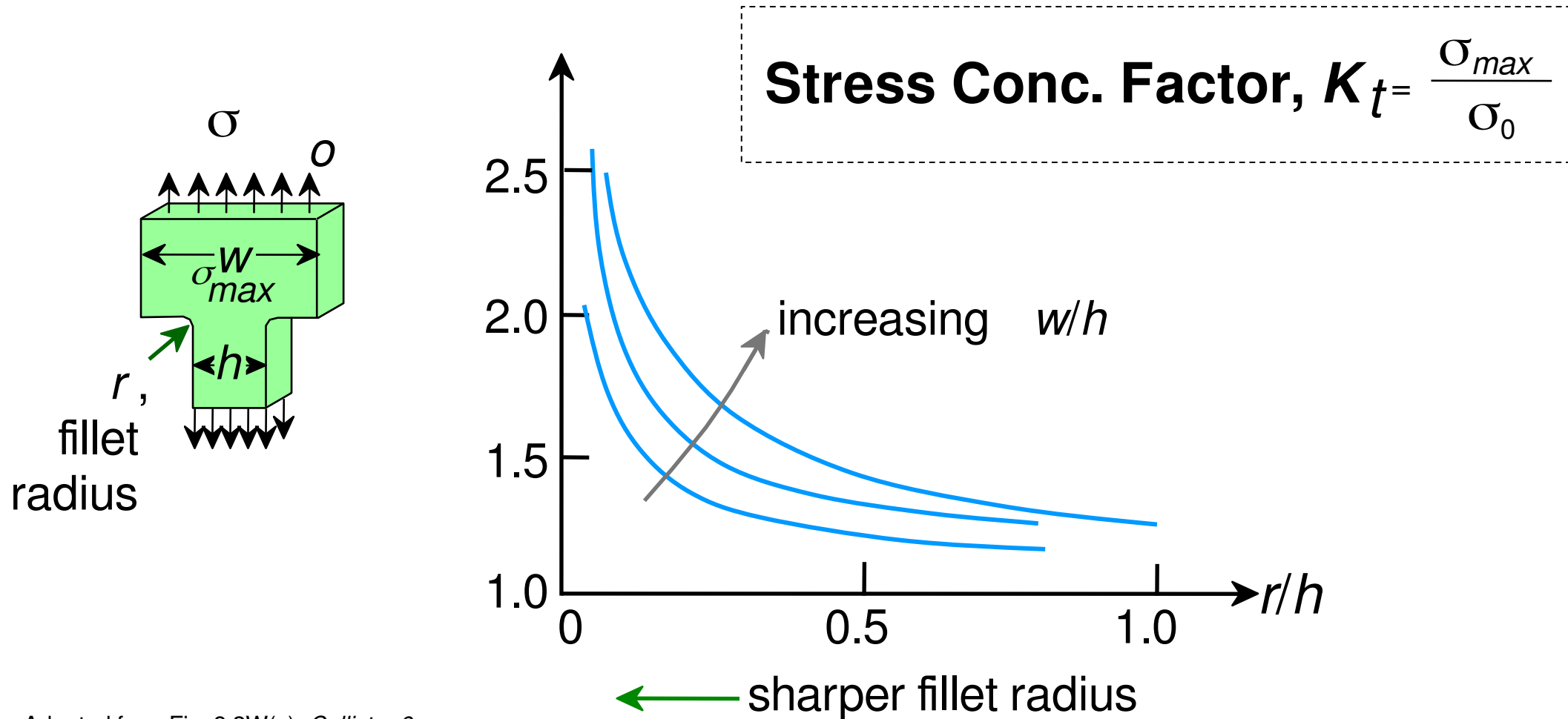
ρ_t = radius of curvature

σ_o = applied stress

$\sigma_{\max}$ = stress at crack tip

Adapted from Fig. 9.8(a), *Callister & Rethwisch 3e*.

Engineering Fracture Design - Avoid sharp corners!

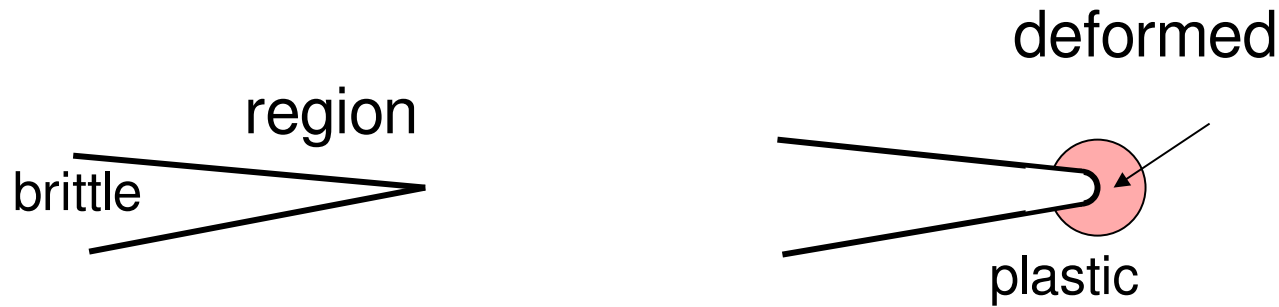


Adapted from Fig. 8.2W(c), *Callister 6e*.
(Fig. 8.2W(c) is from G.H. Neugebauer, *Prod. Eng.* (NY), Vol. 14, pp. 82-87 1943.)

Crack Propagation

Cracks propagate due to sharpness of crack tip

- A plastic material deforms at the tip, “blunting” the crack.



Energy balance on the crack

- Elastic strain energy-
 - energy stored in material as it is elastically deformed
 - this energy is released when the crack propagates
 - creation of new surfaces requires energy

How do you stop a tearing chip bag?



Design Against Crack Growth

- Crack growth condition:

Fracture Toughness $K = Y\sigma_o\sqrt{\pi a} \geq K_{IC}$

- Largest, most stressed cracks grow first!

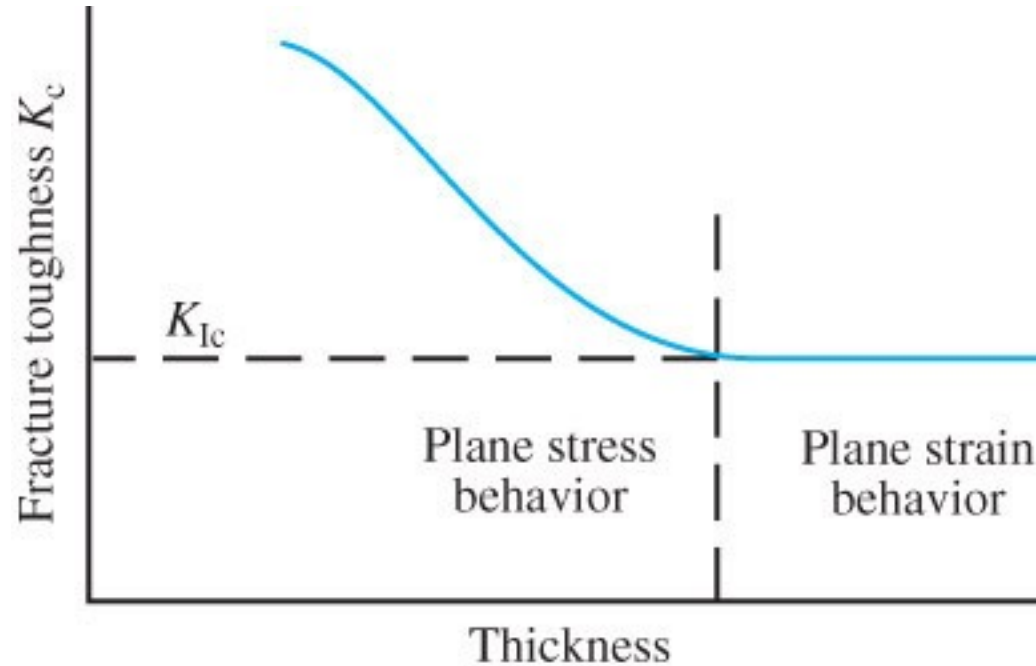
K_{IC} – plain strain fracture toughness (material property)

Y – dimensionless parameter (usually $Y \sim 1$)

σ_o – applied stress

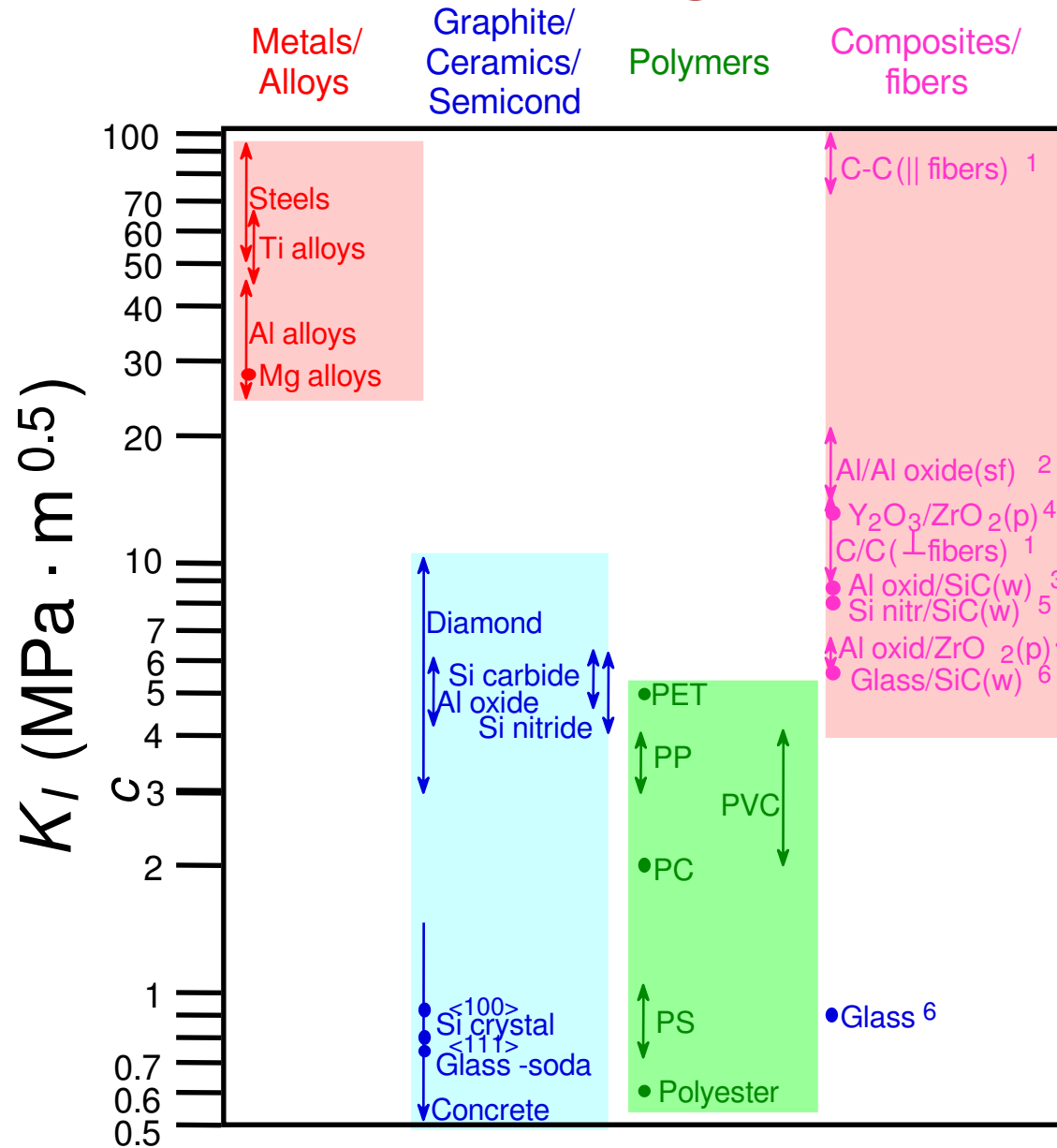
a – length for a surface flaw, ($2a$ for an internal flaw)

Plane Strain Fracture Toughness, K_{IC}



The fracture toughness of a material changes as the geometry (i.e., thickness-to-width) changes until a point is reached where plane strain prevails and the value of K levels off at K_{IC} , a property of the material independent of the geometry.

Fracture Toughness



Example:

An aircraft wing component is constructed from an aluminum alloy that has a plane strain fracture toughness K_{IC} of 26 MPa√m.

If the plate has an internal crack of 18 mm, what is the maximum tensile stress that could be safely applied?

Assume a value of 1.0 for Y .

Example

The value of K_{IC} for a glass optical fiber is $0.75 \text{ MPa}\sqrt{\text{m}}$. The strength of the glass fiber is 69 MPa .

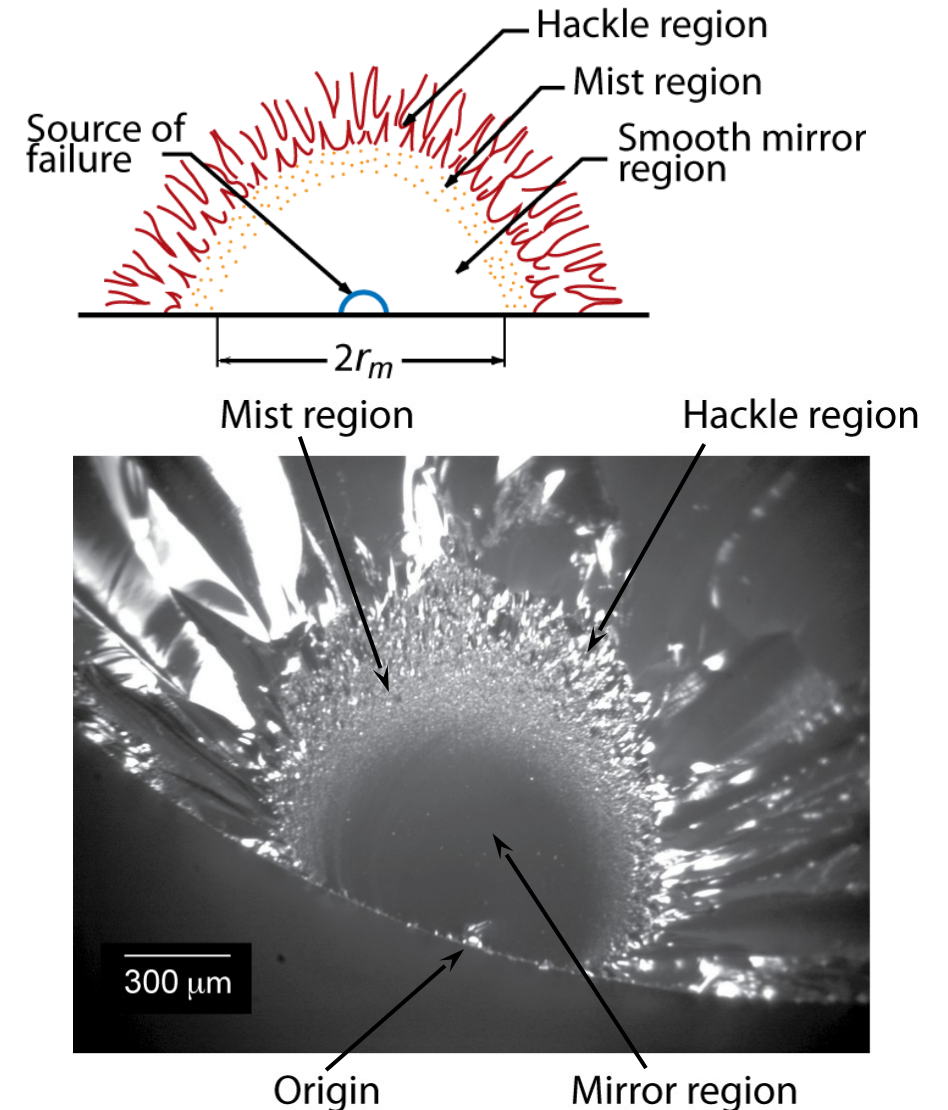
What is the largest surface flaw that could safely exist in such a fiber?

Assume a value of 1.0 for Y .

Brittle Fracture of Ceramics

Characteristic Fracture behavior in ceramics

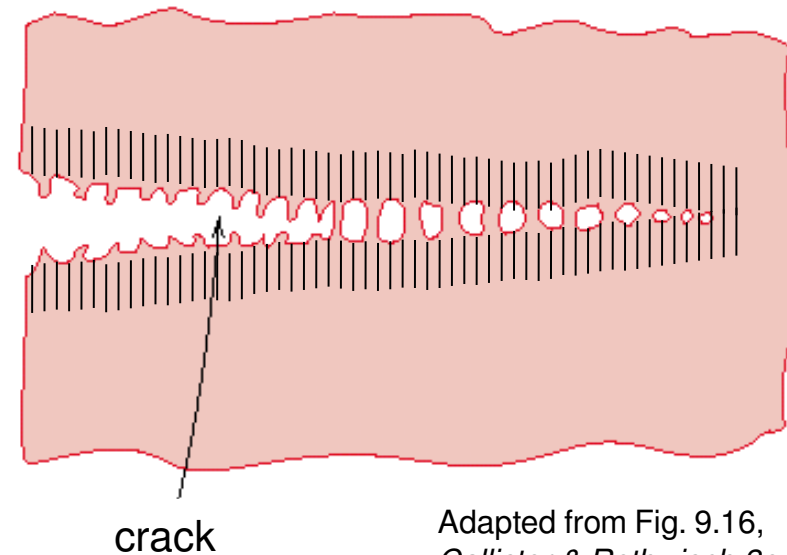
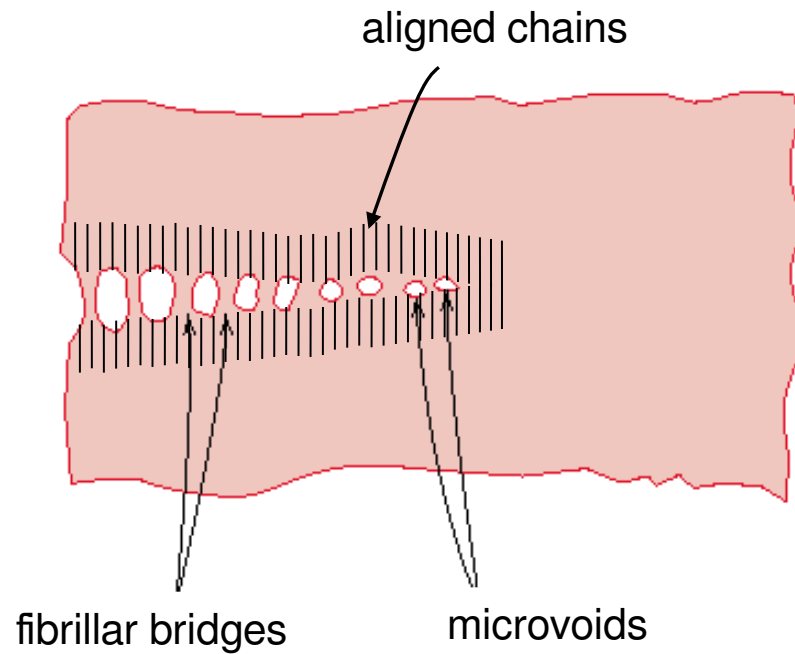
- Origin point
- Initial region (mirror) is flat and smooth
- After reaches critical velocity crack branches
 - mist
 - hackle



Crazing During Fracture of Thermoplastic Polymers

Craze formation prior to cracking

- during crazing, plastic deformation of spherulites
- and formation of microvoids and fibrillar bridges



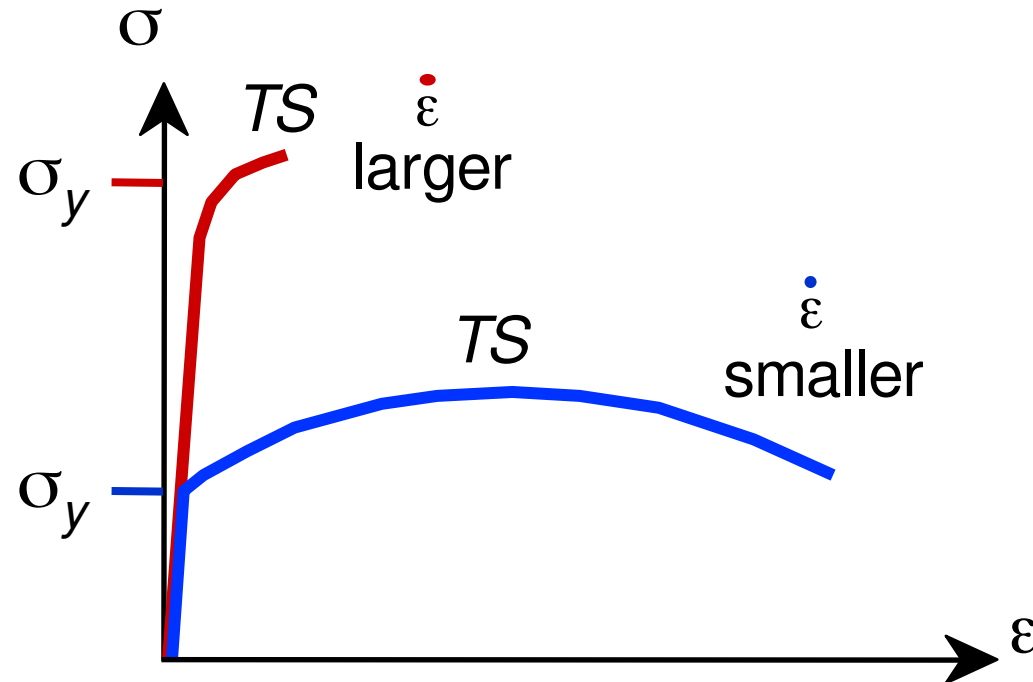
Adapted from Fig. 9.16,
Callister & Rethwisch 3e.

Loading Rate

- Increased loading rate...
 - increases σ_y and TS
 - decreases % EL

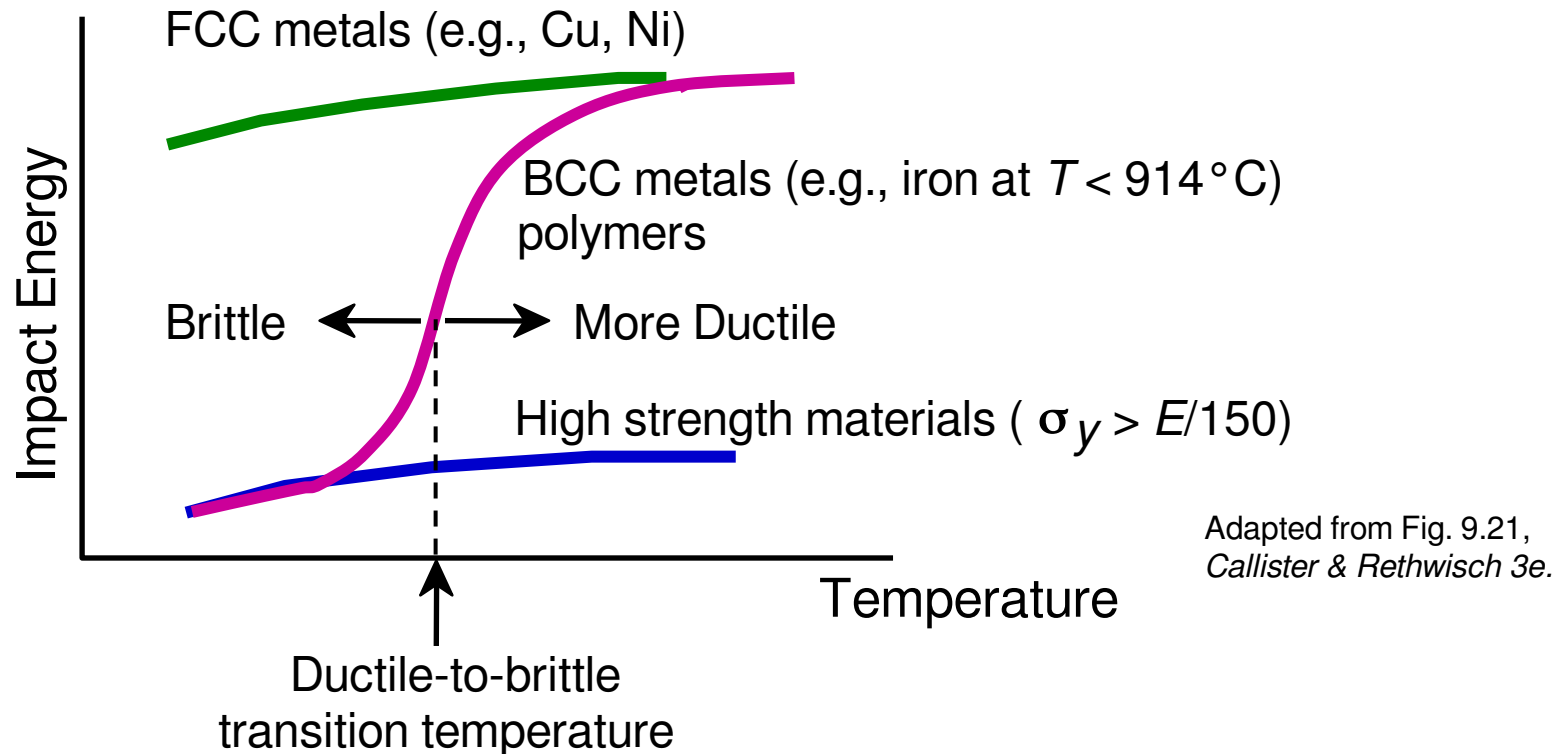
Why?

An increased rate gives less time for dislocations to move past obstacles.



Temperature

- **Increasing temperature...**
 - increases % EL and K_C
- **Ductile-to-Brittle Transition Temperature (DBTT)...**



Field Trip to Lab

- **Module 5 Lab - Rolling**